

Insighta

Project Requirements

A Non-Life Insurance AI
Complaint Ticketing System
(ITS122L-BM1)

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Submitted to:
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I. Introduction

A. Background of the client

Many banks and insurance companies receive a huge number of customer complaints and feedback through many different channels, including email to customer service, online forms, and customer service desks. Many of these complaints pertain to sensitive issues regarding access to accounts, insurance claims, bill payment, and delays in receiving services. Malayan Insurance Philippines is no exception.

In general, most organizations continue to use manual or rule-based ticketing systems that require human staff members to read complaints, categorize them, assign them to staff members, and then resolve them. This is a time-consuming, error-prone, and inefficient way for organisations to process a large amount of unstructured text data from customers. Many times, organisations do not send urgent complaints through the right channels, recurrent issues go unnoticed, and many of the systems set up by organisations for processing complaints are not reliable and are difficult to navigate for many customers.

Moreover, as there is a lack of sophisticated data analysis capabilities, management does not have a comprehensive view of trending complaint issues, customer sentiment, operational weaknesses, and therefore will miss out on the possible opportunities to improve customer service, comply with industry regulations, and protect the loyalty and trust of their customers. The absence of automation, data analysis, and strong security controls eventually leads to the risk increase of data breach and unauthorized access to customer information, which poses a lot of threats not only to the customer but also to the company.

B. Services offered

The designed AI-based Non-life Insurance Complaint Ticketing System will be a SaaS Tool (Software as a Service) to provide a place for customers to submit complaints and give feedback on their experience with the bank insurance company. Complaints that are being submitted through the tool will automatically be converted into support tickets in real-time, giving the customer service team easy tracking, categorization, prioritization, and how long it will take for the issues to be resolved.

The complaint ticketing system takes advantage of AI Technology to analyze the text of customer complaints. Through AI Technology customers can submit complaints based on the type of concern (e.g., Billing, Claims and Account Issues), assess the level of urgency, and prioritize which complaints need to be completed first. This reduces the time that customer service representatives need to spend on handling complaints while allowing them to focus on resolving the more urgent complaints faster.

The Management Dashboard will provide companies with information on how many complaints were submitted for each category, the amount of time it took to resolve a customer's complaint, and how satisfied customers felt about the customer service department after the resolution of their complaints.

Artificial Intelligence Features

- (NLP) Natural language processing can assist in extracting keywords, understanding customer intent and classifying complaints by type.
- (Negative, Neutral and Positive) Sentiment analysis allows us to understand if a complaint is positive, neutral, or negative, as well as assign levels of urgency to each complaint.
- A trend analysis can show us what types of complaints occur on a regular basis, as well as recurring trends in service-related issues.

Cybersecurity Features

- The system has a role-based access control, such as Anonymous, Customer, Staff, and Admin, that needs Secure Authentication
- Data storage and transmission go through a process of Data Encryption for security and privacy
- Logs and Activity monitoring are present to detect suspicious or unauthorized access within the system

Application Type

The system will be developed as a secure web application that is accessible through a web browser. The interface will be designed to be user-friendly, ensuring accessibility for both customers and staff.

II. Problem Statement & Objectives

The business aims to achieve the following objectives by implementing the AI-based Non-life Insurance Complaint Ticketing System:

A. Improve complaint processing speed and operational efficiency

Automating complaint submission, classification, and prioritizing customer complaints eliminates manual processing, error rates, and significantly improves processing time for customer complaints.

B. Increase customer satisfaction and trust in company services

Providing a source of truth via a transparent complaints tracking system will serve as an incentive for customers to place complaints as well as track their resolutions. This will build confidence within the customers about the service delivery and also the response by the business.

C. Support data-driven decision-making for management

Analytical reporting from this system will give management information necessary to determine the volume and type of complaints handled, the response time to complaints registered, and customers' perceptions about the service delivered to them.

D. Ensure compliance with data protection and insurance regulations

In adopting genuine authentication procedures and restricting who can see it to only the staff who need it, the company adheres to all state and federal secrecy laws for data protection.

E. Identify recurring service issues and improve service quality

By studying service problems and weaknesses, it will be possible for the company to address them and thus meet the state and federal regulations.

Table 1. SDG Relevance

SDG	Relevance
SDG 9: Industry, Innovation, and Infrastructure	The system introduces AI-driven automation and NLP into insurance complaint management, promoting innovation in a traditionally manual industry process.
SDG 16: Peace, Justice, and Strong Institutions	By providing transparent complaint tracking, audit logs, and role-based access, the system promotes accountability, institutional transparency, and access to justice for customers.
SDG 8: Decent Work and Economic Growth	Automating repetitive complaint classification tasks allows staff to focus on higher-value resolution work, improving operational efficiency and service quality.

III. Scope of the Project

A. In Scope

- User Authentication & Role Management – Login, logout, session handling, account recovery, and role-based redirection (Customer, Staff, Admin).
- Complaint & Feedback Submission – A unified form for both guests and registered customers to submit complaints or feedback with category selection, description, and optional file attachments.
- Ticket Generation & Tracking – Automatic unique ticket number generation, status lifecycle management (Under Review → In Progress → Pending Customer Response → Resolved → Closed), and tracking views for customers and guests (via one-time links).
- AI Analysis Module – NLP-based text analysis for sentiment detection (Negative/Neutral/Positive), intent identification, issue type classification, and priority assignment (Low/Medium/High).
- Staff Ticket Handling – Staff dashboard for viewing assigned tickets, updating statuses, adding internal notes, and triggering customer notifications.
- Feedback Collection – Post-resolution feedback form with satisfaction rating and optional comments.
- Admin Dashboard & Reports – Analytics dashboard showing complaint volume, trends, resolution time metrics, and sentiment summaries with PDF/CSV export.
- Complaint Category Management – Admin ability to insert, update, activate, or deactivate complaint categories without affecting existing tickets.
- Access Control & Activity Logging – Role-based permissions, guest access rules, and system activity logs for audit and compliance.

B. Out of Scope

- Customer account registration (self-service sign-up) – accounts are assumed to be pre-provisioned.
- Live chat or real-time messaging between staff and customers.
- Mobile application development (the system is web-based only).
- Integration with third-party insurance policy management or CRM systems.
- Payment processing or financial transactions.
- Multi-language support.

IV. Software Requirements: Functional

A. Authentication / Login

(Login & Account Module)

- The system shall allow Registered Customers, Staff, and Admin to log in using verified credentials:
 - Email address
 - Password
 - Confirmed password
 - First name
 - Last name
- The system shall validate credentials and deny access for invalid login attempts.
- The system shall provide a “logout” function that terminates the session securely.
- The system shall restrict access to Staff and Admin pages unless the user is authenticated and authorized under the role-based access control.
- The system shall provide an account recovery mechanism for authorized users.

B. Complaint Ticket Creation

(Complaint & Feedback Submission Module)

- The system shall provide a unified online interface for customers to submit complaints and feedback.
- The system shall require users to specify the ticket type to allow proper classification and routing.
 - The ticket types will be given as:
 - Complaint and
 - Feedback.
- The system shall allow customers to select complaint categories and attach relevant details.
 - Complaint category
 - Billing
 - Claims
 - Account
 - Others
 - Complaint Description
 - Optional Attachments
- The system shall:
 - Automatically generate a unique ticket number
 - Record submission date and time
 - Set initial ticket status to “Under Review”

- If required fields are missing, the system shall prevent submission and prompt the user to complete the form.

C. Ticket Generation, Management, and Tracking

(Ticket Tracking Module)

- The system shall automatically generate a unique ticket number for every submitted complaint.
- The system shall store complete details about the complaint, such as its details, timestamps, status, priority, and timestamps in the database.
- The system shall allow customers to view the current status of their submitted tickets.
 - Guests are given one-time links for ticket status review.
- Ticket statuses shall be the following:
 - Under Review,
 - In Progress,
 - Pending Customer Response,
 - Resolved, and
 - Closed.

D. AI-Based Classification and Prioritization

(AI Decision Support Module)

- The system shall use Natural Language Processing (NLP) in order to classify complaints based on detected intent and issue type.
- The system shall perform sentiment analysis.
 - Negative
 - Neutral
 - Positive
- Sentiment analysis will be used to determine urgency from complaints in order to assign a higher priority level.
 - Low
 - Medium
 - High

E. Customer Service Operations

(Staff Ticket Handling Module)

- The system shall allow the Staff role to view and manage assigned tickets.
- The system shall allow the Staff role to update ticket status.
- The system shall notify the customer of ticket updates.

F. Complaint List View

- If the user is a customer, the system shall display only tickets submitted by that customer.
- If the user is staff or admin, the system shall display all tickets based on priority and status.
- The system shall allow filtering by status and category.

G. Customer Feedback Collection

(Feedback Module)

- The system shall prompt customers to provide feedback after a ticket is resolved.
- The system shall allow customers to rate satisfaction and submit comments.
- The system shall store feedback for reporting and analysis.

H. Management Dashboard and Reporting

(Reports & Dashboard Module)

This function allows management to monitor system performance.

- The system shall generate analytical dashboards and ready-to-download reports.
- The system shall display complaint trends, resolution times, and sentiment summaries with appropriate visualizations.
- The system shall allow the management to review the historical reports.

I. Security and Access Control

(Access Control & Logs Module)

- The system shall enforce role-based access to ensure security and administration.
 - Access roles will be given as (1) Customer, (2) Staff, and (3) Admin.
 - Customer User Types will involve Registered and Guest (Unregistered).
 - The system shall allow Guest users to submit complaints or feedback without requiring an account login.
 - The system shall require Guest users to provide a valid email address in order to receive ticket updates.
- The system shall require secure authentication for all authorized users.
- The system shall log system access and activity.

V. Software Requirements: Non-Functional

A. Performance

- The system shall process and generate complaint tickets in near real time.

B. Security

- The system shall ensure that sensitive information of customers is encrypted at all instances of storage and transmission.
- The system shall comply with applicable data protection laws.

C. Usability

- The system should provide a user-friendly web interface accessible via modern web browsers.

D. Availability

- The system shall be available during the standard business hours with minimal downtime.

E. Reliability & Data Integrity

- The system shall ensure, in the case of an unexpected error, that submitted complaints and ticket updates are saved reliably and never lost.
- The system shall maintain audit logs of ticket status changes and user actions to support traceability and integrity.

F. Scalability

- The system should support increasing volumes of submitted tickets without major performance degradation.
- The system shall support multiple staff users accessing and updating tickets simultaneously.

G. Accessibility

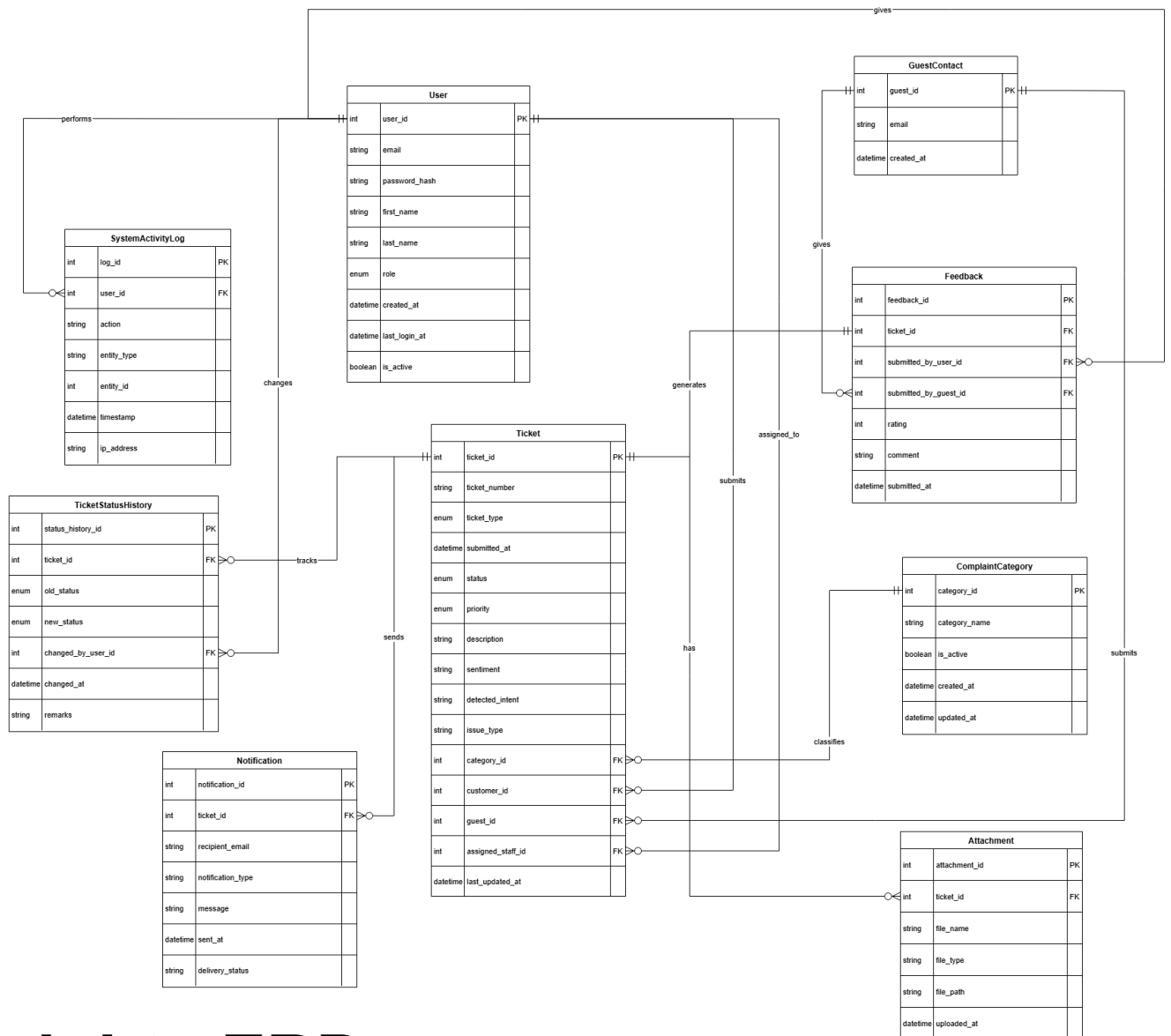
- It shall implement basic principles of accessibility on the Web (e.g., providing readable text and good form labels).

H. Maintainability

- The system should be designed in modules (ticketing system, analysis system using AI, dashboard, authenticate system) so that it could be easily updated.

- The system should enable administrators to update categories for complaints without having an impact on already recorded tickets.

VI. Database Design



Insighta ERD

Figure 1. Insights Database Design

The system database is structured around **9 core entities** that support the full complaint ticketing lifecycle:

Table 2. Insighta Core Entities

Entity	Purpose
User	Stores registered users (Customer, Staff, Admin) with credentials and role
GuestContact	Stores guest email and contact info for unregistered submitters
Ticket	Central entity holding complaint/feedback details, status, priority, sentiment, and AI fields
ComplaintCategory	Lookup table for complaint categories (e.g., Billing, Claims, Account) with active/inactive status
Attachment	Stores uploaded files linked to a ticket
TicketStatusHistory	Audit trail of all status changes per ticket with timestamps and user who made the change
Feedback	Post-resolution satisfaction rating and comments linked to a ticket
Notification	Records of notifications sent to users on ticket updates
SystemActivityLog	Logs of login attempts, user sessions, and system-level actions for audit purposes

VII. Features of the System

The interaction requirements from the participants are founded upon the current and future interaction, which is related to the business purpose and the place of interaction.

A. Filing Complaints and/or Feedback

- The complaint/feedback is formalized by the customer using this system whenever they have any problem or comments related to the insurance services.
 - This would include a category indicating whether this input is a complaint or feedback.
- Additionally, the customer would also identify the category of the complaint made, which would enable the system to classify them correctly so that they may be quickly resolved.
- Along with complaints, the system shall allow customers to submit the general feedback in order to collect customer suggestions, compliments, and general concerns that will be utilized for the reports and monitoring.

B. Ticket Creation and Tracking

- Using this system, complaints lodged will be automatically generated as numbers for following up and addressing by the customer services team in the best way possible.
- Customers are able to monitor their reported tickets in order to track the status of the complaint.

C. AI-Based Complaint Analysis

- This system will rely on Artificial Intelligence (AI) in order to help determine the intention behind the complaint, type, and urgency levels of the complaint.
- The customer service agents shall receive support tickets that have been prioritized by AI to make them attend to only the critical issues at hand.

D. Complaint Resolution

- Customer service staff shall use a customer complaint ticket that has been allotted to them to investigate and contact, and resolve in the most efficient way, the customer complaint.
- Members of staff would be able to change the status of the ticket in order to provide current status information to the client concerning the matter they reported.

E. Feedback and Satisfaction Monitoring

- Feedback will also be sought for customers subsequent to resolving a complaint, aimed at creating customer satisfaction and offering customers a chance to have improvements brought to their attention regarding their delivery of excellent customer service.

F. Management and Reporting

- The management team will employ the help of analytical dashboarding in order to evaluate and manage complaints, the time taken to solve complaints, customer feelings towards a solution, and operational vulnerabilities.
- Patterns in customer complaints will be analyzed by management in order to determine where they might improve service and maintain regulatory requirements.

G. Security and Access Control

- Users of the system will be classified as Guest (Unregistered), Customer (Registered), Staff, or Admin, each with varying access levels to system features and information.
 - Guest users shall be allowed to submit complaints or feedback without creating an account.
 - Customers shall have access to additional features, including ticket history and status tracking.
 - Staff shall update ticket statuses and cater to customer concerns and feedback.
 - Admin shall oversee the entire system, including monitoring and further analysis for insurance service improvement.
- This will prevent accidental access to and loss of confidential information by ensuring that all authorized users have secure authentication.

VIII. Wireframes

A. Customer (on an Iphone 17 Pro Max)





INSIGHTA



Thank you, Name!

We've received your concern. Our team is working on it and will get back to you as soon as possible

Ticket ID: #AB12345



Track My Complaint

[FAQs](#)

[Privacy Policy](#)

[Contact Us](#)



INSIGHTA

Enter Ticket ID

Please input your Ticket ID for verification and to track your complaint



Enter Your Ticket ID

#AB12345

NEXT

[FAQs](#)

[Privacy Policy](#)

[Contact Us](#)



INSIGHTA

Track My Complaint

See the status of your complaint.

Ticket ID: #AB12345

Submitted on February 5, 2026
at 11:20 AM

Category: Billing Issue



In Progress



Current Status

Your complaint is being reviewed by our support team.

Estimated time of Resolution

1-3 business days



Need Help?

Have any additional questions?

CONTACT US

[FAQs](#)

[Privacy Policy](#)

[Contact Us](#)



INSIGHTA

Contact Us

We're here to assist you, feel free to reach out.



Call Us

(032) 2569889

Monday - Saturday, 8:00 AM - 6:00 PM



Email Us

insighta@gmail.com

We'll respond as soon as possible



Visit Us

#123 Financial Center, Makati City

Monday - Saturday, 8:00 AM - 6:00 PM

[FAQs](#)

[Privacy Policy](#)

[Contact Us](#)



INSIGHTA

Privacy Policy

Your privacy is important to us. This policy explains how we collect, use, and protect your information when you use our services.



Information We Collect

We collect information you provide to us directly, as well as data collected automatically when you use our services.



How We Use Your Information

We use your information to provide, maintain, and improve our services, communicate with you, and ensure security.



How We Protect Your Data

We implement security measures to protect your information from unauthorized access, alteration, or disclosure.



Your Privacy Rights and Choices

You have rights regarding your personal data, such as accessing, correcting, or deleting your information.

If you have any questions or concerns about our Privacy Policy, please don't hesitate to contact us.

CONTACT US

[FAQs](#)

[Privacy Policy](#)

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INSIGHTA

Track My Complaint

See the status of your complaint.

Ticket ID: #AB12345

Submitted on February 5, 2026
at 11:20 AM

Category: Billing Issue



Complete



Current Status

Your issue has been resolved!

Time of Resolution

February 6, 2026 - 11:42 AM



Your issue has been successfully resolved! If you need any further assistance or have additional questions, please feel free to reach out anytime. A feedback from you will be appreciated!

CONTACT US

RATE US

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B. Admin





ADMIN

TICKETS

Dashboard

Tickets

Reports

Profile

All

Open

In Progress

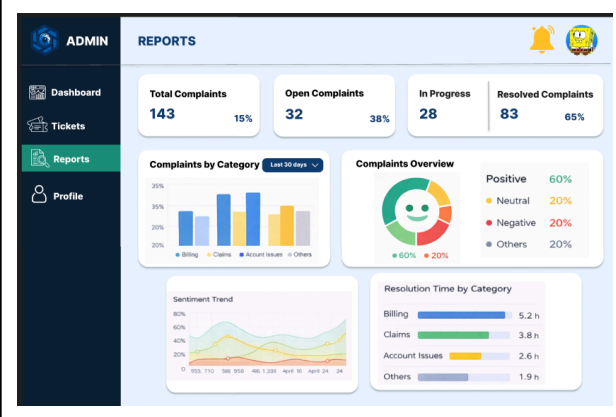
Resolved

Search ticket...

Filter by

+ NEW TICKET

Ticket ID	Category Type	Customer	Status	Priority	Agent	Date
#AB12345	Billing	Sally Dale Cruz	Open	High	Kristy Nicolas	2028-02-02
#AB67899	Account Issue	Ruff Zabala	In Progress	Medium	Jela Nicolas	2028-01-22
#AB10112	Claims	Denz Manning	Resolved	Low	Ann Nicolas	2028-01-06



ADMIN

PROFILE

Dashboard

Tickets

Reports

Profile

Jela Nicolas

Admin

Full Name

Annjela Nicolas

Email:

aknics@gmail.com

Role:

Admin

Username:

Jela Nicolas

Phone Number:

09937322863

Employee ID:

AKN14356

EDIT PROFILE

CHANGE PASSWORD

CHANGE PICTURE

LOGOUT

IX. Client Proof of Agreement

February 18, 2026

Malayan Insurance Philippines
Integrated Marketing Communications
ETY Building, Quintin Paredes, Binondo, Manila

Dear Mr. Guanzon:

We hope this letter finds you well. We are Computer Science students of **Mapúa University** under the course **ITS122L/BM1 - Web Systems and Technologies 2 Laboratory**. As part of our course requirements, we are proposing to develop "**Insighta: A Non-Life Insurance AI Complaint Ticketing System**" for **Malayan Insurance Philippines**.

The proposed system is a secure, web-based ticketing platform where customers may submit complaints and feedback, which are then converted into trackable tickets. It includes ticket status tracking, staff handling workflows, and a management dashboard for reporting and monitoring. The system also includes AI-assisted analysis, such as intent/issue classification and sentiment-based prioritization, to support faster complaint handling.

Data Privacy Notice

To ensure confidentiality and avoid any privacy risk, this project will **NOT** require, request, store, or process any real customer/company data from your organization.

All testing, demonstrations, and reports **will use ONLY mock or synthetic data** (such as simulated complaint messages, dummy ticket records, and fictional user accounts) to demonstrate the end-to-end workflow. **NO production database access, NO live integrations, and NO real policyholder information will be involved at any stage.**

Our group will handle the full development lifecycle, including requirements implementation, database design, UI development, and system testing based on the agreed scope and specifications.

If you have any questions, concerns, or suggestions regarding the project scope, we are open to clarifications and feedback to ensure our deliverables match expectations within the academic timeline.

With this, we respectfully request your **approval/acceptance** to proceed with the development of the project for academic purposes.

Thank you for your time and consideration.

Sincerely,

Cloie Sofia Dela Cruz
3rd Year Mapúa University Student

Eugene Carl Dela Cruz
3rd Year Mapúa University Student

Vincent Andrei Lusung
3rd Year Mapúa University Student

Vincent Piamonte
3rd Year Mapúa University Student

Ralf Cedric Zabala
3rd Year Mapúa University Student


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